



Biopolyesters Diligence for a Sustainability Focused VC

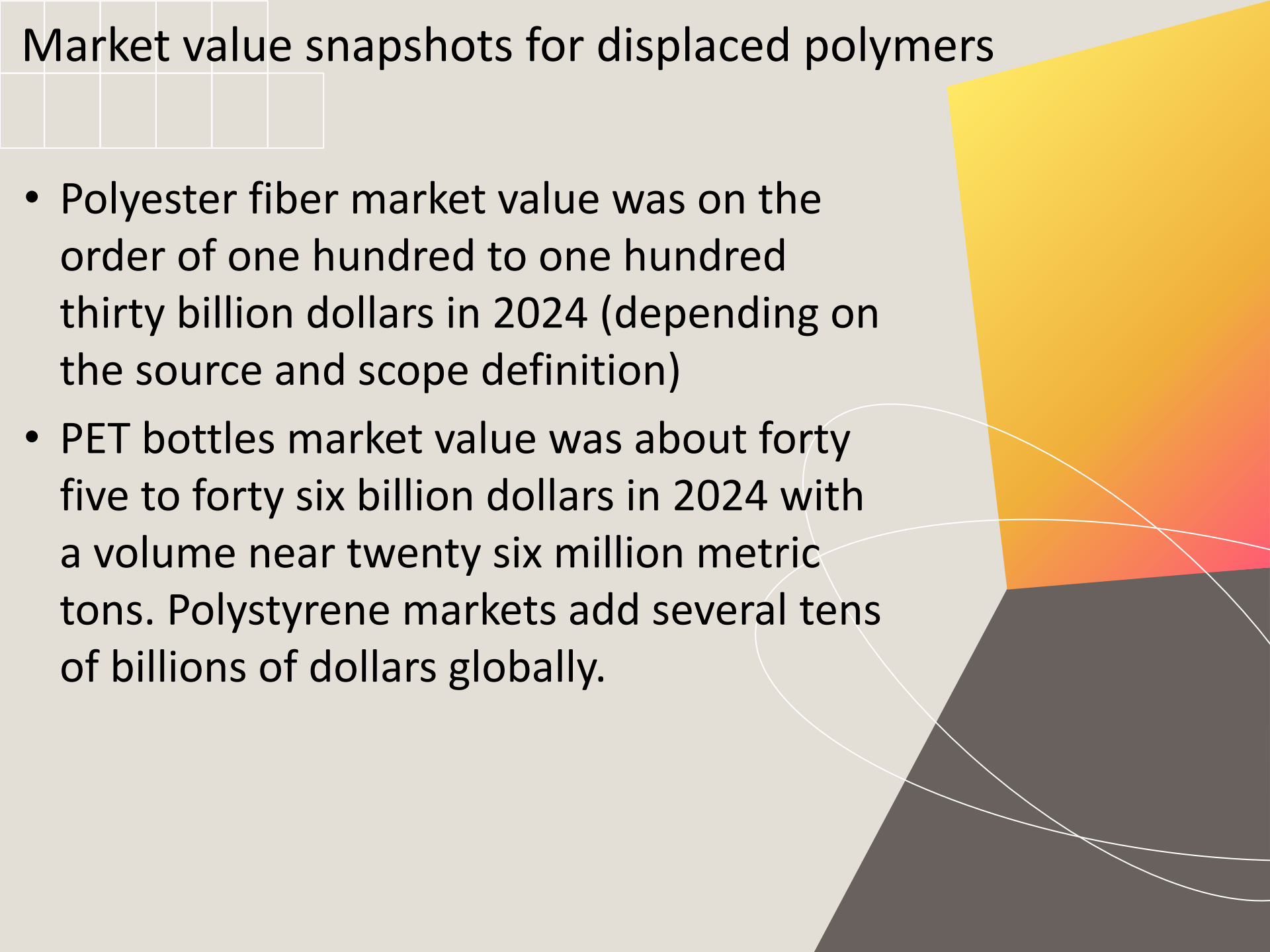
Which plastics could bio polyesters replace and how large are those markets?

- Rigid packaging can include beverage bottles, thermoformed clamshells, and many food containers. The incumbent materials are PET and polystyrene, and some polypropylene in injection molded parts.
- Flexible packaging can include produce bags and liners. Polyhydroxyalkanoates can substitute for polyethylene or polypropylene when certified compostability is required.
- Fibers and textiles can include apparel and nonwovens: polyester dominates, while polylactic acid fibers are niche and growing in specific segments.
- Medical and specialty items can include resorbable implants, sutures, pins, and agricultural mulch films where biodegradation or resorption has actual value.
- Polyester fiber was about seventy one million metric tons in 2023 and about seventy eight million metric tons in 2024. PET bottles are about twenty six million metric tons. global plastics production was roughly four hundred to four hundred thirty six million metric tons in 2023.
- Bioplastics capacity in 2024 was about two point four seven million metric tons with actual production about one point four four million metric tons, which signals meaningful headroom if the demand materializes.

Examples of items that can be made from bio polyesters

- Polylactic acid can be used for thermoformed food packages, cold cups and lids, coffee capsules, and 3D printing filament. It also appears in a lot of medical devices that benefit from resorption.
- Polyhydroxyalkanoates can be used for compostable straws and cutlery, compostable bags, and agricultural films. Some grades actually show faster biodegradation in certain test environments.

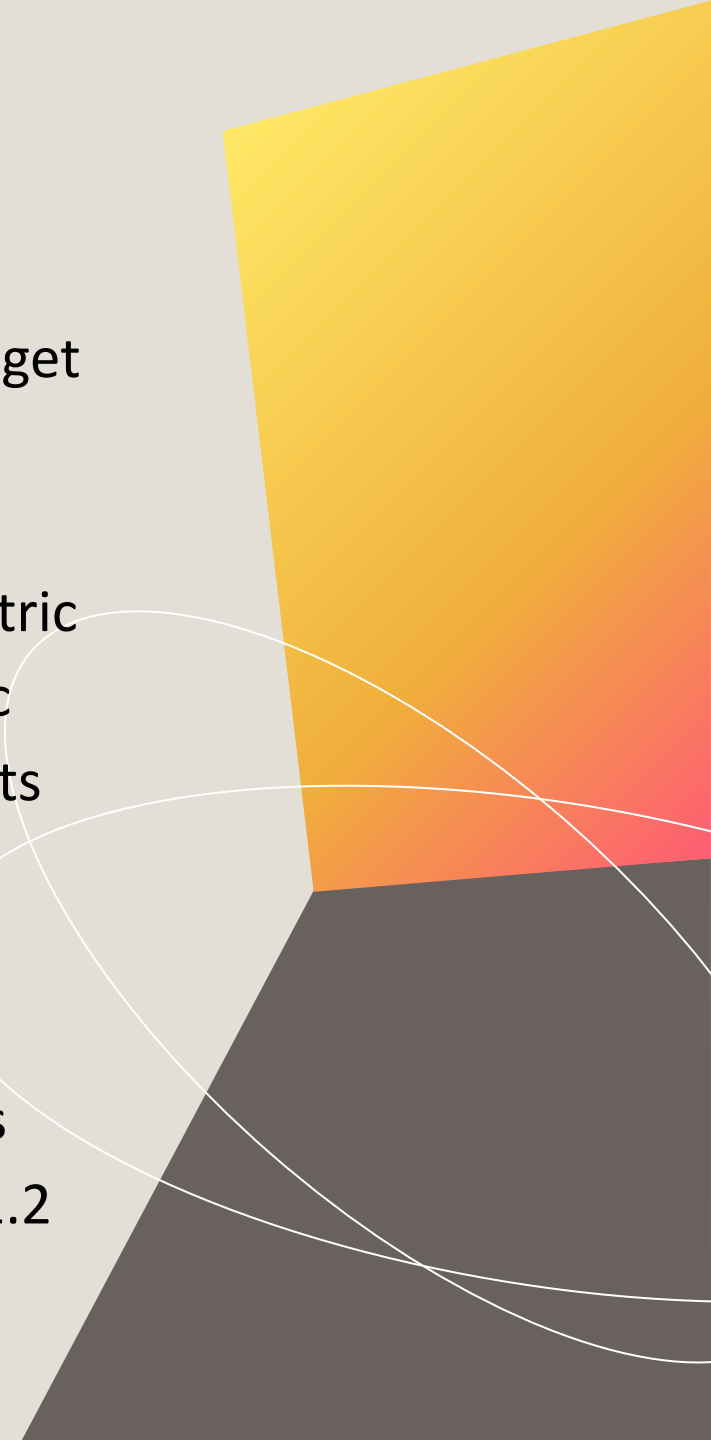
Market value snapshots for displaced polymers



- Polyester fiber market value was on the order of one hundred to one hundred thirty billion dollars in 2024 (depending on the source and scope definition)
- PET bottles market value was about forty five to forty six billion dollars in 2024 with a volume near twenty six million metric tons. Polystyrene markets add several tens of billions of dollars globally.

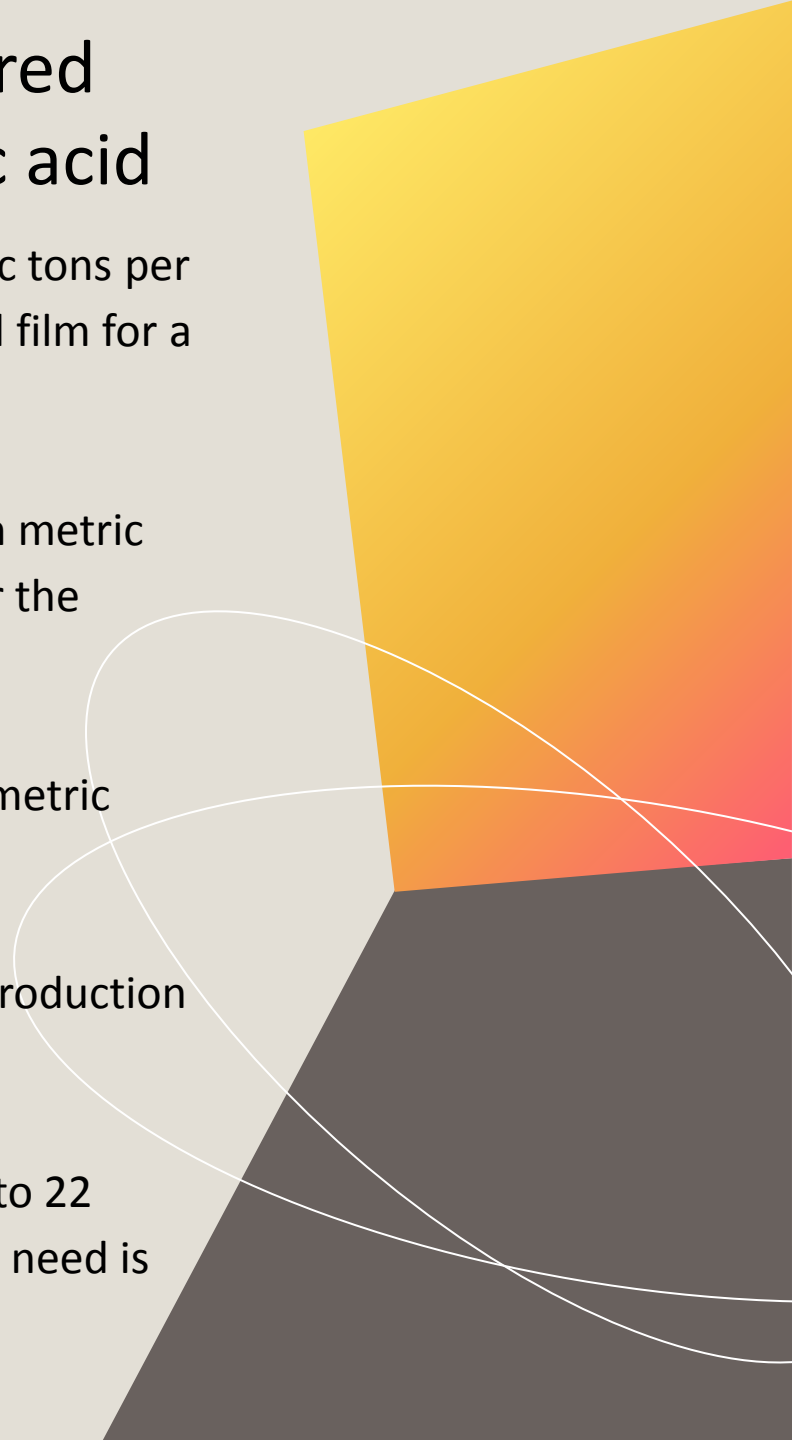
Assumptions to convert corn into polylactic acid

- Fermentation turns glucose into lactic acid at close to a one to one yield. In practice, plants get very near that result. The lactic acid is then polymerized into polylactic acid pellets.
- Studies suggest you need about 1.6 to 2.5 metric tons of corn to make 1 metric ton of polylactic acid. The exact number depends on coproducts and how the plant is integrated, so we use a range instead of a single estimate.
- For land use, we can think about it as like two simple yardsticks. In the United States, yield is about 179 bushels per acre, which is about 11.2 metric tons per hectare. The global average is about 6.05 metric tons per hectare.



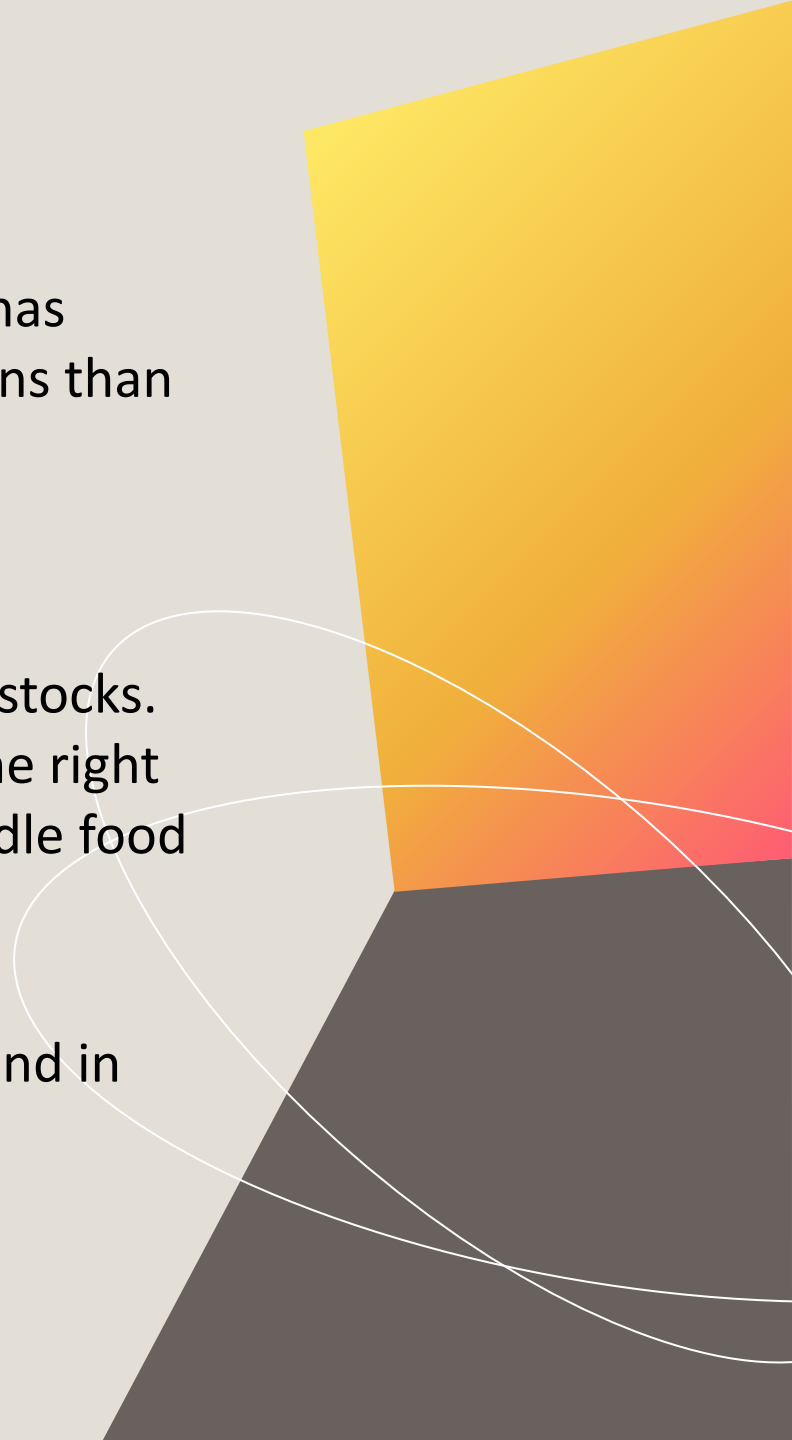
Corn and land to replace one hundred percent of polyester with polylactic acid

- The target polyester market is about 100 million metric tons per year when you add up polyester fiber, PET bottles, and film for a simple first pass.
- The polylactic acid needed would be about 100 million metric tons, and this ignores property limits that would lower the realistic near-term substitution rate.
- The corn required would be about 162 to 250 million metric tons.
- That equals about 13% to 20% of current world corn production if global output is about 1.23 billion metric tons.
- Using a United States yield, the land need is about 14 to 22 million hectares. Using a global average yield, the land need is about 27 to 41 million hectares.



Sustainability advantages to highlight

- Many studies find that polylactic acid often has lower cradle to gate greenhouse gas emissions than polystyrene or PET. The results change a lot depending on the place and power source.
- Bio based carbon cuts reliance on fossil feedstocks. When the product and the region support the right end of life route, organics programs can handle food soiled items that don't recycle too well.
- Some PHA grades break down faster in soil and in water.

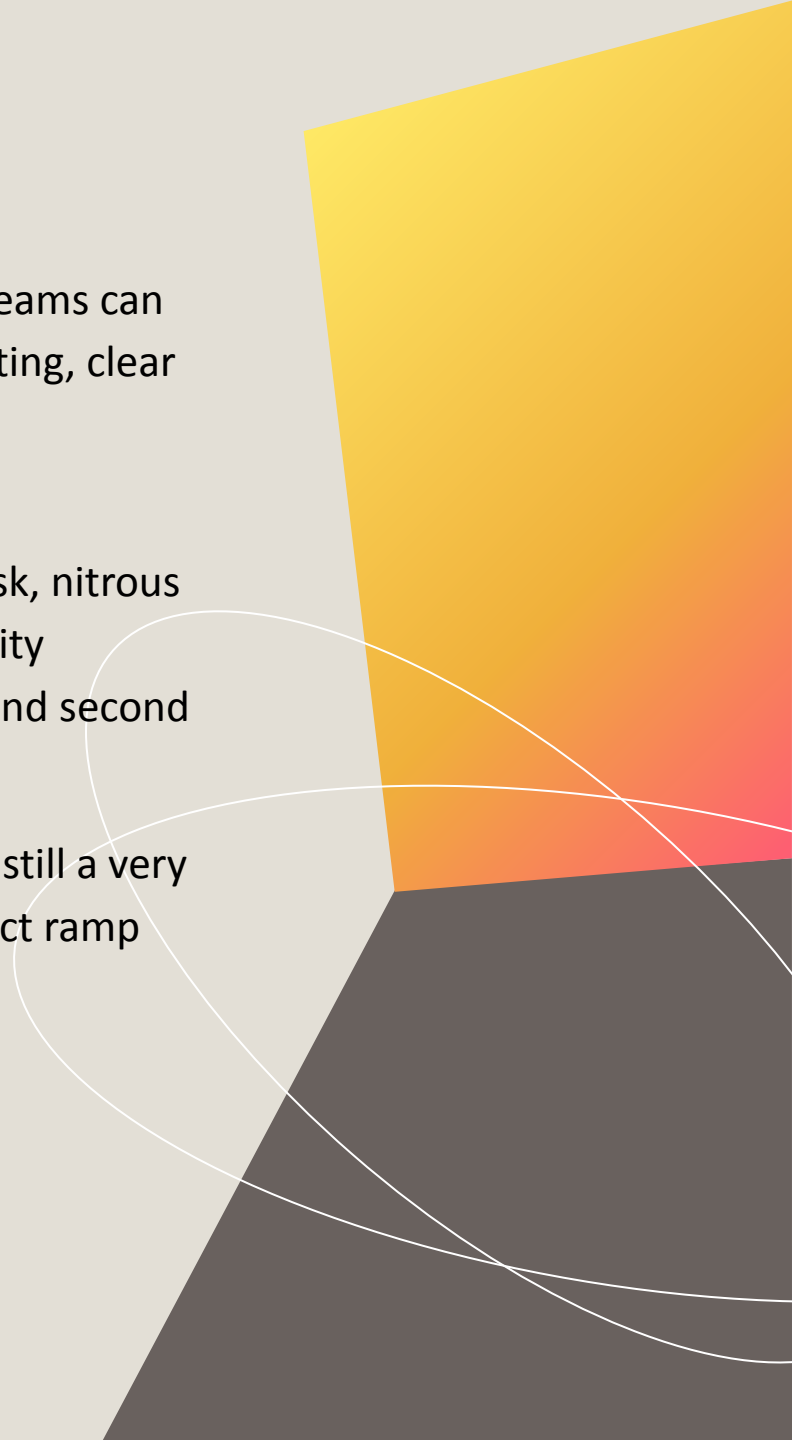


Sustainability disadvantages and operational risks

Mixing polylactic acid into polyethylene terephthalate streams can lower recycled resin quality. Facilities will need better sorting, clear markers, and policy aligned labels to keep streams clean.

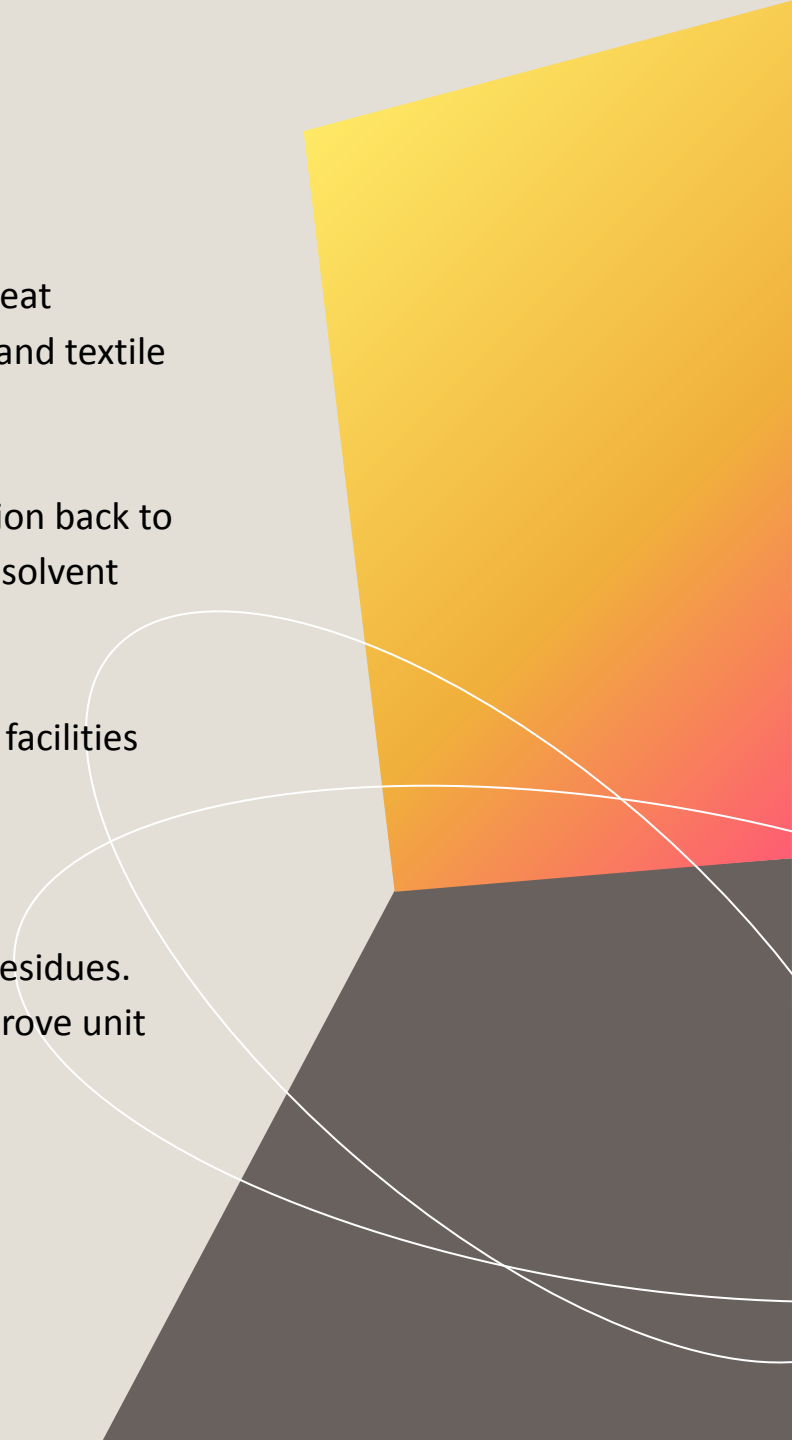
Corn based supply chains have a lot of land use change risk, nitrous oxide from fertilizer, irrigation water needs, and biodiversity impacts. Better farming practices and a shift to residues and second generation feedstocks can reduce these impacts by a lot.

Scale and cost remain constraints because bioplastics are still a very small share of total plastics. Investors would have to expect ramp risk and a lot of price volatility.



Areas where a start up could create value

- Materials performance: use additives and copolymers to raise heat resistance, toughness, and barrier strength, so more packaging and textile uses accept the material without losing too much quality.
- End of life technology: develop low temperature depolymerization back to lactide or lactic acid with high yield and low energy. Build some solvent routes that allow a lot of recycle cycles with little property loss.
- Sorting and labeling: create markers and digital systems that let facilities identify polylactic acid and polyhydroxyalkanoates at scale.
- Feedstocks and process: shift to second generation sugars and residues. Raise fermentation productivity and lower nutrient costs to improve unit economics.



Citations

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Thank you